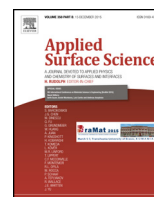




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Effect of iron content on the structure and mechanical properties of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ and $(\text{AlTi})_{60-x}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_x$ ($x=15, 20$) high-entropy alloys



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ARTICLE INFO

Article history:

Received 23 April 2015

Received in revised form 29 June 2015

Accepted 30 July 2015

Available online 1 August 2015

Keywords:

Multicomponent alloys

Mechanical properties

Casting

Scanning electron microscopy

X-ray diffraction

ABSTRACT

In this work, we investigated the microstructure and mechanical properties of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ high entropy alloys, produced by arc melting and casting in an inert atmosphere. The structure of these alloys was studied by X-ray diffractometry and scanning electron microscopy. The as-cast alloys were heat treated at 773, 973 and 1173 K for 1800 s to investigate the effects of aging on the plasticity, hardness and elastic properties. Compared to the conventional high-entropy alloys the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys are relatively hard and ductile. Being heat treated at 973 K the $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ alloy shows considerably high strength and relatively homogeneous deformation under compression. The plasticity, hardness and elastic properties of the studied alloys depend on the fraction and intrinsic properties of the constituent phases. Significant hardening effect by the annealing is found.

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1. Introduction

There are many high temperature applications where metallic alloys with superior mechanical and functional properties are required. A recently proposed concept of multi-component high-entropy alloys (HEA) consisting of refractory elements [1–3] opened up interesting possibilities in the alloy developments satisfying such demands. These HEAs are equiatomic (but not always) multicomponent alloys, wherein configurational entropy is maximized, leading the solid solubility and formation of a single or a dual-phase structure [4–6]. Such solid solutions with multiprincipal components are found to be in a metastable state due to the sluggish diffusion of atoms [7–9], preventing intermetallic compounds from precipitating. It is widely accepted [10] that the formation of high entropy refractory alloys by casting is facilitated by small atomic size difference and moderately negative enthalpy of mixing between the constituent elements, which is opposite to bulk metallic glasses [11] having relatively large values.

The promising properties of high entropy alloys offer the potential to be used in many applications, such as tools, molds, dies, mechanical parts and furnace parts, which require high strength, thermal stability as well as wear and oxidation resistance, with application temperatures up to 800 °C.

The present study was taken up to prepare three new equiatomic, multi-component HEAs. We have investigated the effects of Fe addition element on the structure, mechanical properties and thermal stability of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys. As thermal stability is an important factor for industrial application of HEA it is also worthwhile to investigate the effect of annealing on the structure and properties evolution. The mechanical properties the effects of annealing on the structure were also investigated in the present work.

2. Experimental procedure

The ingots with nominal compositions $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ were prepared by induction melting using mixtures of the high-purity elements (>99.9 at.% purity) in an argon atmosphere purified by Ti-getter. Cylindrical rod samples with a diameter of 3 mm were obtained

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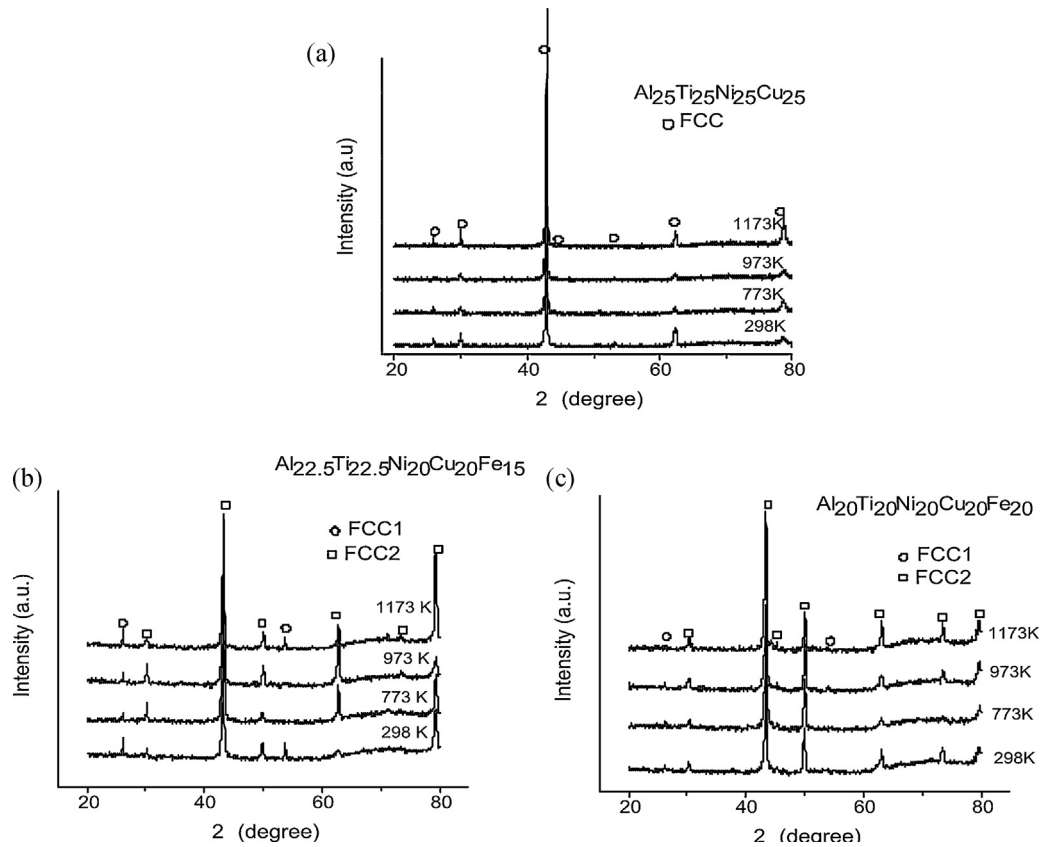


Fig. 1. X-ray diffraction pattern of the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ (a), $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ (b) and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ (c) alloys in the as-cast and annealed states. Relationship between Δ and ΔH_{mix} for equiatomic multi-component alloys (equiatomic amorphous phase forming alloys (AM); non-equiatomic amorphous phase forming alloys; solid solution phases (SS) and intermetallic compound (IC)) [16].

by remelting the master alloy in a quartz tube and subsequent injection into a copper mold in an inert gas atmosphere.

In order to examine the thermal stability of the samples, isochronal annealing was performed at 773 and 1173 K for 30 min in a vacuum and under argon atmosphere of 2×10^{-2} Pa.

The structure of the alloys was examined at cross-sections of the as-cast samples by X-ray diffraction (XRD) with Bruker D8 diffractometer with $\text{Cu K}\alpha$ radiation. The morphologies of the surface of these specimens subjected to mechanical tests were observed using a Hitachi S-4800 Scanning Electron Microscope (SEM).

In order to evaluate the mechanical properties the cylindrical specimens with an aspect ratio of 2:1 were uniaxially compressed at room temperature until fracture (using a Shimadzu testing machine with 50 kN maximum load). The specimens for compression testing were cut from the 3 mm rods and carefully polished to have a final length of 6 mm, while ensuring that their ends planes were parallel to each other. The compression test was performed according to the American Society for Testing and Materials (ASTM) standard (E9-09), at an initial loading rate of $5 \times 10^{-4} \text{ s}^{-1}$. Vickers microhardness was measured on polished cross-section surfaces using a HM211 Mitutoya device.

3. Results and discussion

3.1. Theory analysis and discussion

In this paper the enthalpy of mixing, atomic-size difference and a parameter Ω for the multi-component $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$,

Table 1

Calculated heat of mixing, mixing entropy, melting temperature, atomic mismatch and Ω parameter of the refractory HEA alloys.

Alloys	ΔH_{mix} (kJ/mol)	ΔS_{mix} ($\text{J K}^{-1} \text{mol}^{-1}$)	T_m (K)	Ω	δ
$\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$,	−23.25	14.4	1490	0.92	7.16
$\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$	−19.8	13.3	1536	1.00	7.15
$\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$	−17.6	13.37	1554	1.18	7

$\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys were calculated using the following equations [12,13]:

$$\Delta H_{\text{mix}} = \sum_{i=1, i \neq j}^n \Omega_{ij} c_i c_j \quad (1)$$

$$\Delta S_{\text{mix}} = -R \sum_{i=1}^n c_i \ln c_i \quad (2)$$

$$\Delta S_{\text{mix}} = R \ln N \quad (3)$$

where $\Omega_{ij} (= 4\Delta H_{\text{AB}}^{\text{mix}})$ is the regular melt-interaction parameter between i th and j th elements, and $\Delta H_{\text{AB}}^{\text{mix}}$ is the mixing enthalpy of binary liquid alloys, the value can be obtained in ref. [10–13].

$$\Delta(\delta) = \sqrt{\sum_{i=1}^n c_i (1 - r_i/\bar{r})^2} \quad (4)$$

here n is the number of the components in an alloy system, c_i is the atomic percentage of the i th component, $\bar{r} (= \sum_{i=1}^n c_i r_i)$ is the

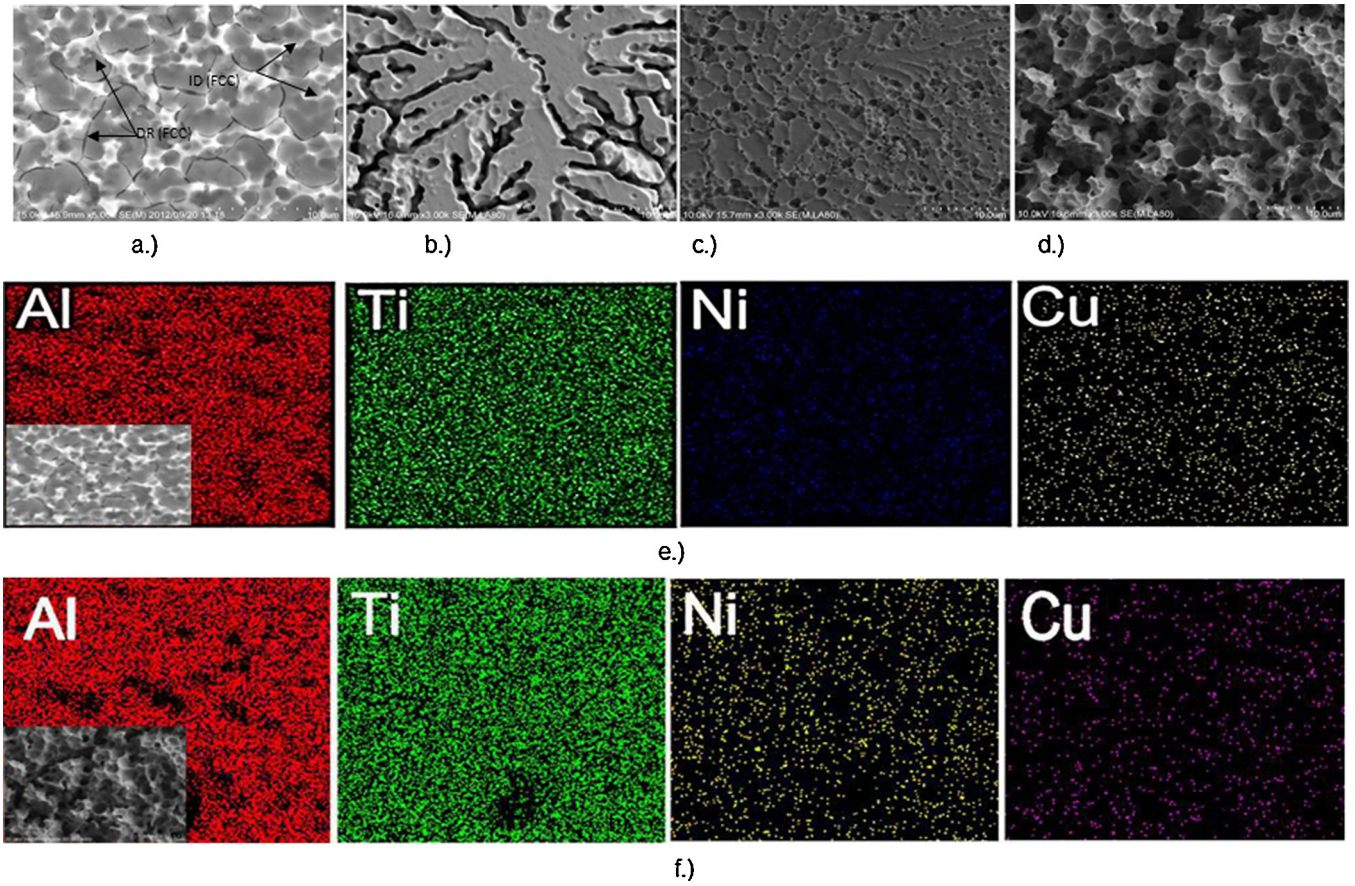


Fig. 2. SEM image of the surface of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ HEA in as-cast state (a) and as-annealed at 773 (b), 973 (c), 1173 K (d), (e) and (f) are a EDS map of alloying components.

average atomic radius, and r_i is the atomic radius which could be obtained in Ref. [14,15].

The parameter Ω is defined for predicting the solid-solution formation for various multi-component alloys as:

$$\Omega = \frac{T_m \Delta S_{\text{mix}}}{|\Delta H_{\text{mix}}|} \quad (5)$$

The melting temperature of n-elements alloy, T_m , is calculated using the rule of mixtures:

$$T_m = \sum_{i=1}^n c_i(T_m)_i \quad (6)$$

Here, $(T_m)_i$ is the melting point of the i th component of alloy. As analyzed above, the value of Ω is positive, and $\Omega=1$ should be proposed as a critical value to form a solid solution. If $\Omega > 1$, the contribution of $T\Delta S_{\text{mix}}$ will exceed that of ΔH_{mix} for a solid-solution formation, and the multi-component HEAs are mainly composed of the solid-solutions; if $\Omega \leq 1$, ΔH_{mix} is the predominant part in the equation for, and the formation of an intermetallic compound(s) is expected to prevent the formation of the solid-solution phases in multi-component HEAs [11]. In the case of $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloy $\Omega=1.18$. Small atomic size difference and the value of Ω slightly larger than 1 indicates that the strain energy caused by lattice distortion is lower in $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloy and the entropy effect of ΔS_{mix} is large enough to exceed that of ΔH_{mix} for the solid-solution formation. Thus the component atoms can randomly distribute in the alloy and a simple solid solution is stable and easily forms during solidification [15].

Calculating the ΔH_{mix} , δ data for a number of HEA's and metallic glasses, Guo and Liu [14] see Fig. 1 could delimitate the

following regions in ΔH_{mix} versus δ diagram: HEA with a solid solution phase form when the three parameters are in the suitable range: $-22 \leq \Delta H_{\text{mix}} \leq 7$ kJ/mol, and $\delta < 8.5$ and $\Delta S_{\text{mix}} \leq 19.5$ J/(K mol). In the case of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys the atomic-size difference $\delta=7.16$, 7.15, 7, the mixing enthalpy $\Delta H_{\text{mix}} = -23.25$, -19.8 , -17.6 kJ/mol and $\Delta S_{\text{mix}} = 14.4$, 13.3, 13.37 J/(K mol), respectively, as can be seen Table 1.

The values of Ω and δ for $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys correspond to those forming a solid solution. For the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ alloy the mixing enthalpy is large enough and intermetallic phases precipitate. However, it is surprised to note that in the region where solid solutions form, intermetallic phases can also form. The mixing enthalpy and entropy is not the only factor that controls the solid solution formation in equiatomic multi-component alloys. The critical that determines the formation of solid solution phases is the atomic size difference (δ). In the case of the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ alloy a solid solution can be formed owing to the small atomic mismatch ratio [14,15].

3.2. X-ray diffraction and SEM analysis

Fig. 1a–c shows the X-ray diffraction patterns of the cast and heat-treated alloys. To confirm the thermal stability of the phases, the samples were annealed at 773, 973 and 1173 K for 1800 s. After annealing at 773, 973 K the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ alloy preserves its original FCC1 phase ($\text{Ti}_{24}\text{Ni}_{28}\text{Al}_{67.04}$). The structure of $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys in the as-solidified state contains the FCC1 ($\text{Ti}_{24}\text{Ni}_{28}\text{Al}_{67.04}$) phase and FCC2 ($\text{Cu}_{0.8}\text{Fe}_{0.2}$) [18,19]. This double-phase structure is also

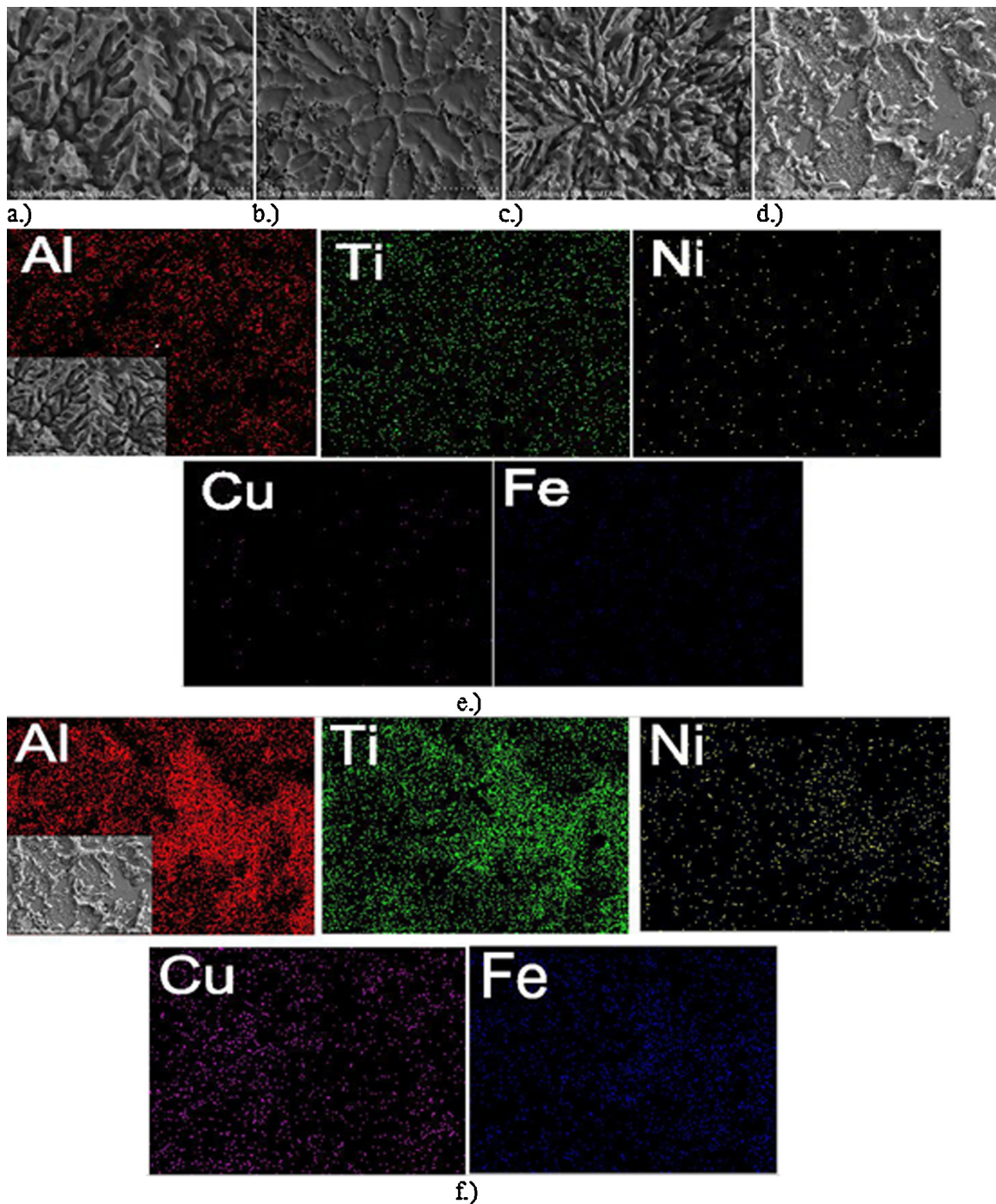


Fig. 3. SEM image of the surface of $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ HEA in as-cast state (a) and as-annealed at 773 (b), 973 (c), 1173 K (d), (e) and (f) are EDS map of alloying components at 773 and 1173 K.

preserved and remained almost unchanged after annealing (Fig. 1(b) and (c)). Comparing the relative X-ray intensity of the strongest peaks of FCC1 and FCC2 phases with that of the base alloys $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$, it can be seen that the amount of Cu and Fe in the FCC1 phase is slightly lower than the average alloy composition which means that the FCC phase rejects Cu and Fe into the grain boundaries and form a FCC2 phase ($\text{Cu}_{0.8}\text{Fe}_{0.2}$).

The lattice parameters determined by the XRD are 356.2, 365.4 and 364.7 pm for the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys, respectively.

Since Al, ($r=146$ pm) and Ti ($r=147$ pm) atoms the biggest of five elements, the lattice constant of the FCC phase decreases as the aluminum content decreases. Moreover, since the size of Fe is the smallest, 124 pm, the lattice constant of the FCC becomes the larger as the content of Fe decreases from 20 to 15 at.%.

Figs. 2–4 show the results of SEM observations of the as-cast and as-annealed after etching in Keller's reagent of the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys, respectively. Typical cast dendrite and interdendrite structures (defined as DR and ID in the figures, respectively) are observed in these as-cast and annealed alloys.

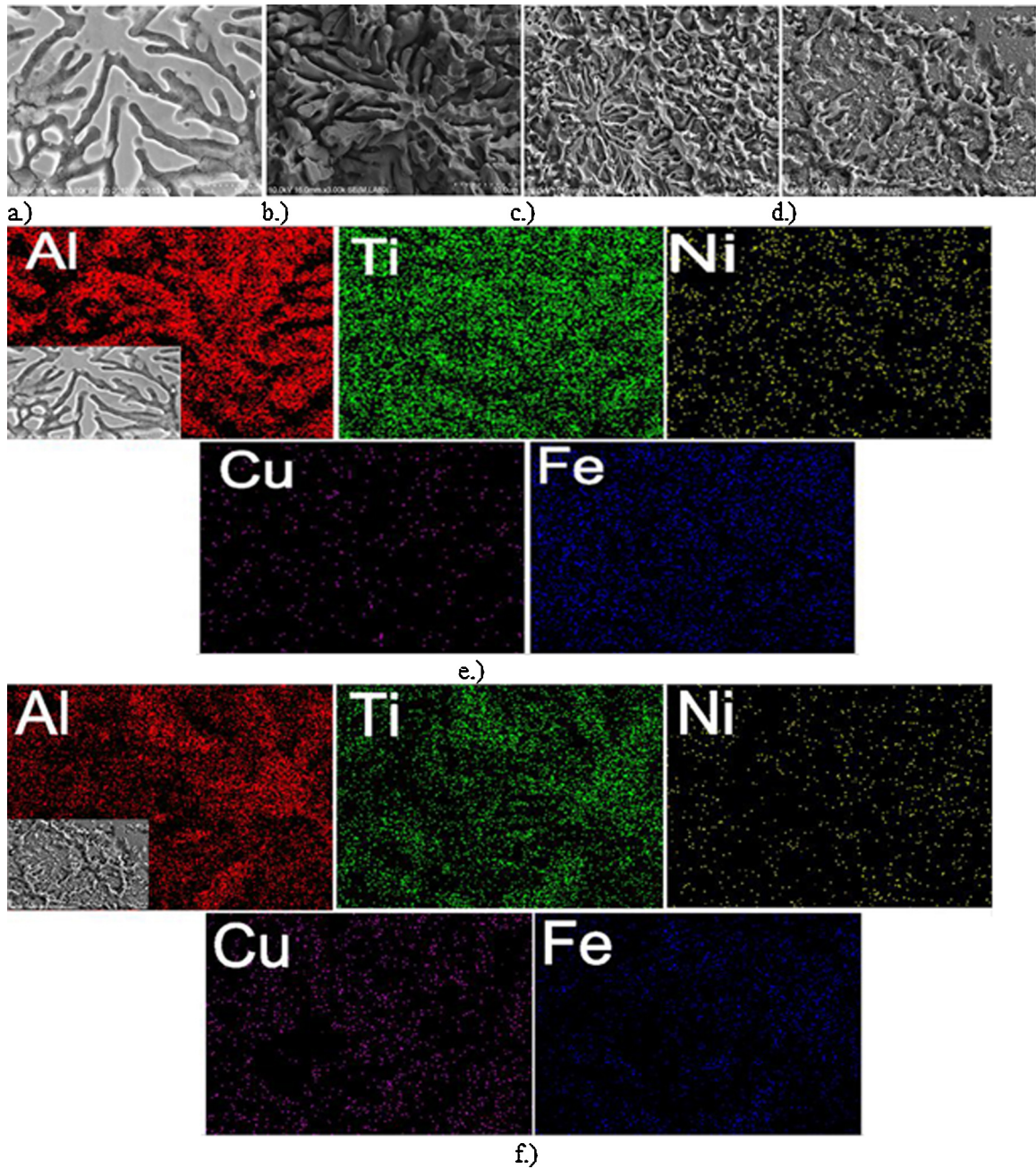


Fig. 4. SEM image of the surface of $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ HEA in as-cast state (a) and as-annealed at 773 (b), 973 (c), 1173 K (d), (e) and (f) are EDS map of alloying components.

It can be deduced from the XRD results that the dendritic phase must be the FCC1 solid solution structure while the interdendritic phase must be the FCC2- $\text{Cu}_{0.8}\text{Fe}_{0.2}$.

However, the morphology of dendrite and interdendrite is diverse in this alloy system indicating a difference in the annealing temperature (973 K) could have a large effect on the growth behavior of dendrites. Cooper segregation to interdendritic space is typically seen in these alloy systems. This can be explained with the bonding energy between cooper and other elements. Since the mixing enthalpy for a mixture of cooper and each of aluminum, titanium, nickel and iron is -1 , -9 , 4 and 13 kJ/mol [12], it leads to the fact that cooper dislikes being placed in dendrite with larger aluminum, titanium, nickel and iron concentration. Combined with the XRD analysis, both the dendrite and interdendritic spaces in

$\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ are of simple FCC phase.

Fig. 5a–c shows the true stress (σ) vs. strain (ϵ) curves of the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ (a), $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ (b) and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ (c) alloys obtained during compression testing for as-cast and heat treated alloys. The yield stress (σ_y), values for the as-cast samples 300, 980 and 1074 MPa, respectively, are given in Table 2.

It should be noted that the plastic strain (ϵ_p) for the $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloy annealed at 973 K exceeds 13.5% during compression test (see Fig. 5(c)), in the case of the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ alloy almost nothing plastic deformability is observed.

As for the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys aged at 973 K, the yield stress and

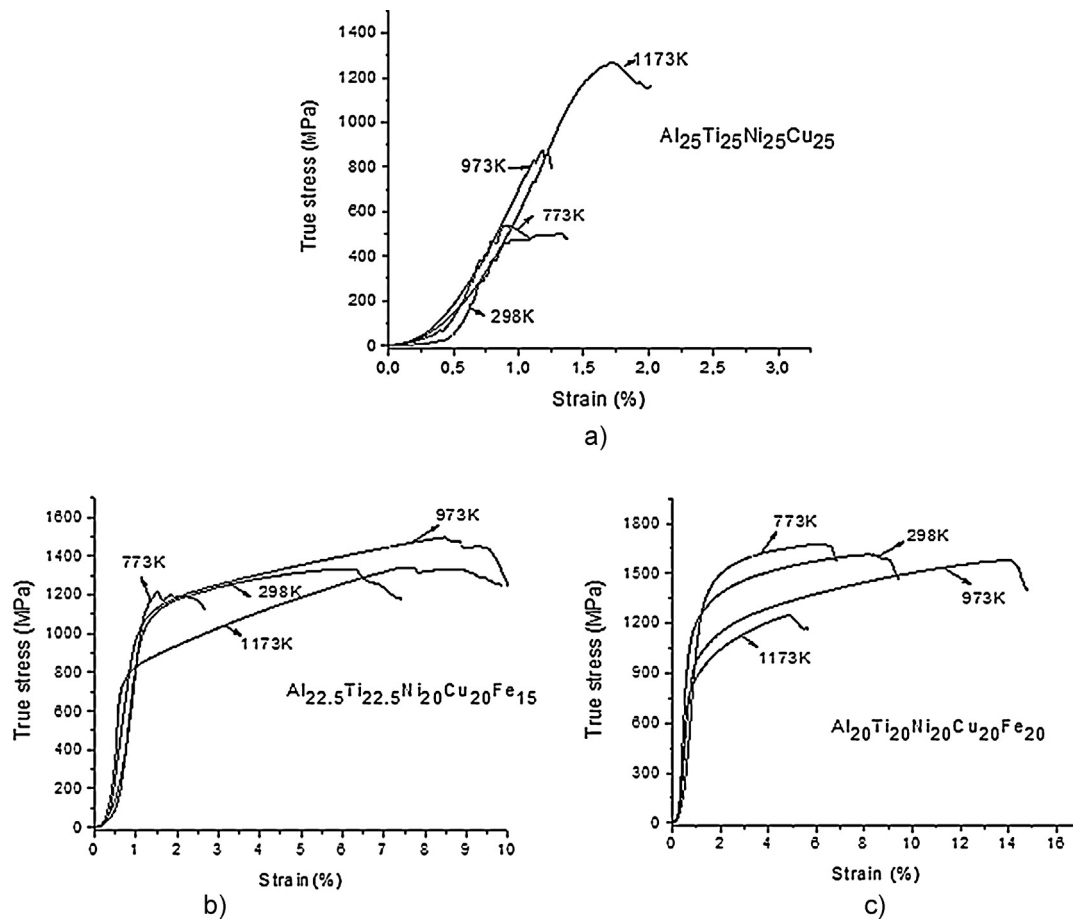


Fig. 5. True stress vs. strain compression curves for the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ (a), $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ (b) and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ (c) alloys in as-cast and as-annealed states.

Table 2

Chemical composition of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys in atomic percentage.

	$\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ 298 K		$\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ 1173 K	
	White at.%	Gray at.%	White at.%	Gray at.%
Al	18.15	19.44	19.72	18.93
Ti	17.05	16.82	17.63	19.72
Ni	26.09	26.47	23.9	23.6
Cu	38.71	37.27	38.75	37.76

	$\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ 298 K		$\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ 1173 K		$\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ 298 K		$\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ 1173 K	
	Dendrite at.%	Interdendrite at.%	Dendrite at.%	Interdendrite at.%	Dendrite at.%	Interdendrite at.%	Dendrite at.%	Interdendrite at.%
Al	12.7	22.32	23.02	20.56	19.78	19.2	20.33	9.08
Ti	24.57	19.84	20.56	19.21	20.05	16.93	16.23	4.76
Ni	26.96	19.09	21.12	23.53	27.62	27.31	23.41	8.73
Cu	20.58	25.27	22.74	24.34	12.64	24.49	27.1	71.4
Fe	15.20	13.48	12.56	12.35	19.91	12.07	12.93	6.03

plastic strain are 760 MPa, 0.96%, 989 MPa, 8.5% and 836 MPa, 13.5% respectively (see Table 2).

In the case of $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys the best combination of the compression mechanical properties is found at 973 K.

Fig. 6 shows the variation in the Vickers hardness of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$, $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys as a function of the annealing temperature. An average of ten measurements obtained for each sample,

on randomly selected places. In the case of $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys the confidence interval value increases with temperature up to 973 K indicating that the structure likely becomes more inhomogeneous. The $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ alloy exhibits the Vickers hardness values higher than those of the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloy. For the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ alloy we can see almost no changes in hardness until 973 K, but when the annealing temperature reached 1173 K one can see drastic changes in it. Such an excellent

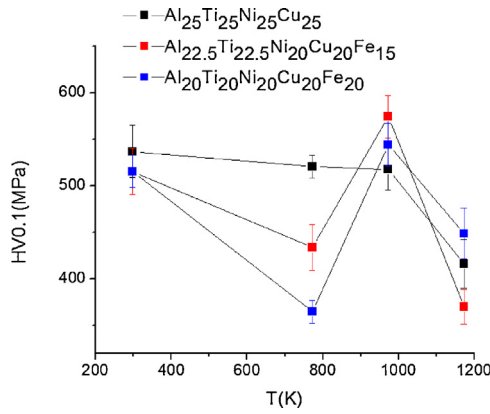


Fig. 6. Vickers hardness of Al₂₅Ti₂₅Ni₂₅Cu₂₅, Al_{22.5}Ti_{22.5}Ni₂₀Cu₂₀Fe₁₅ and Al₂₀Ti₂₀Ni₂₀Cu₂₀Fe₂₀ as-cast and heat-treated alloys measured at room temperature after annealing at T_{ann} for 1800 s.

Table 3

Room-temperature compressive mechanical properties for Al₂₅Ti₂₅Ni₂₅Cu₂₅, Al_{22.5}Ti_{22.5}Ni₂₀Cu₂₀Fe₁₅ and Al₂₀Ti₂₀Ni₂₀Cu₂₀Fe₂₀ as-cast and heat-treated alloys, namely Young's modulus (E), yield stress (σ_y), fracture strength (σ_{max}), plastic strain (ε_p) and Vickers hardness at 1N load.

Alloy	Misrostructure	T _{annealing} (K)	E (GPa)	σ _y (MPa)	σ _{max} (MPa)	ε _p (%)	HV _{0.1}
Al ₂₅ Ti ₂₅ Ni ₂₅ Cu ₂₅	FCC	298	108	300	536	0.85	537
		773		284	498	0.67	521
		973		760	870	0.96	518
		1173		1103	1263	1.2	416
Al _{22.5} Ti _{22.5} Ni ₂₀ Cu ₂₀ Fe ₁₅	FCC1+FCC2	298	145	980	1326	6.6	516
		773		1065	1210	1.8	434
		973		989	1490	8.5	575
		1173		737	1335	8.8	370
Al ₂₀ Ti ₂₀ Ni ₂₀ Cu ₂₀ Fe ₂₀	FCC1+FCC2	298	146	1074	1617	8.2	516
		773		1158	1670	5.8	365
		973		836	1569	13.5	544
		1173		697	1247	4.3	449

resistance to high temperature annealing softening is seldom seen in conventional alloys (Table 3.).

4. Conclusion

Three new refractory alloys namely: Al₂₅Ti₂₅Ni₂₅Cu₂₅, Al_{22.5}Ti_{22.5}Ni₂₀Cu₂₀Fe₁₅ and Al₂₀Ti₂₀Ni₂₀Cu₂₀Fe₂₀, were produced by induction-melting and casting. Only a simple FCC phase was formed in the Al₂₅Ti₂₅Ni₂₅Cu₂₅ alloy as a result of annealing at 573, 773, 973 and 1173 K. Almost no changes were observed in the Vickers microhardness of the Al₂₅Ti₂₅Ni₂₅Cu₂₅ alloy until the annealing temperature of 973 K. At 1173 K it shows a greatly decrease to the hardness value for all three alloys.

FCC1 and FCC2 solid solutions structures appear in the as-cast and as-annealed Al_{22.5}Ti_{22.5}Ni₂₀Cu₂₀Fe₁₅ and Al₂₀Ti₂₀Ni₂₀Cu₂₀Fe₂₀ alloys. Elemental segregation can be observed with the dendrite region enriched in Al, Ti and Ni while the interdendrite region is enriched in Cu and Fe. This kind of alloys exhibits high resistance to annealing softening.

In the case of the Al_{22.5}Ti_{22.5}Ni₂₀Cu₂₀Fe₁₅ and Al₂₀Ti₂₀Ni₂₀Cu₂₀Fe₂₀ alloys a strong decrease followed by a strong increase

in the microhardness as a function of the annealing temperature. Most the alloys in the annealed state possess even higher Vickers microhardness than the as-cast alloys. The Al_{22.5}Ti_{22.5}Ni₂₀Cu₂₀Fe₁₅ and Al₂₀Ti₂₀Ni₂₀Cu₂₀Fe₂₀ alloys annealed at 973 K show the highest compressive stress and ductility values, which is about that for the as-cast samples of 980 and 1074 MPa, respectively.

The CoCrFeNiTiAl_x high entropy alloys possess integrated high temperature mechanical property as well but more expensive than the composition of our alloys [17].

Acknowledgements

E. Fazakas express her gratitude to the Matsumae International Foundation and WPI-Advanced Institute for Materials Research, Tohoku University making possible a 6 month stay in Sendai. This work was supported by World Premier International Research Center Initiative (WPI), MEXT, Japan and by the Ministry of Education and Science of the Russian Federation in the framework of Increase Competitiveness Program of NUST «MISiS» (№ K2-2014-013) and by Hungarian OTKA grants 84078 and 109570.

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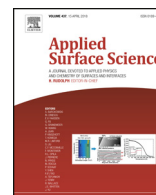
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Update

Applied Surface Science

Volume 437, Issue , 15 April 2018, Page 453

DOI: <https://doi.org/10.1016/j.apsusc.2017.07.228>



Erratum

Erratum to “Effect of iron content on the structure and mechanical properties of $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ and $(\text{AlTi})_{60-x}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_x$ ($x = 15, 20$) high-entropy alloys” [Appl. Surf. Sci. 358 (2015) 549–555]



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ARTICLE INFO

Article history:

Received 19 July 2017

Accepted 24 July 2017

Available online 10 August 2017

The following amendments are necessary:

Page 552: The lattice parameters determined by the XRD for the FCC1 phase are 583.2 pm for the $\text{Al}_{25}\text{Ti}_{25}\text{Ni}_{25}\text{Cu}_{25}$ alloy, 582.3 pm for the $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ alloy and 581.3 pm for the $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloy. The lattice parameters determined by the XRD for the FCC2 phase are 365.4 and 364.7 pm for the $\text{Al}_{22.5}\text{Ti}_{22.5}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{15}$ and $\text{Al}_{20}\text{Ti}_{20}\text{Ni}_{20}\text{Cu}_{20}\text{Fe}_{20}$ alloys, respectively.

Page 553: Cooper should be Copper

We sincerely apologize to the readers for any inconvenience it caused.

DOI of original article: <http://dx.doi.org/10.1016/j.apsusc.2015.07.207>.

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<https://doi.org/10.1016/j.apsusc.2017.07.228>
0169-4332/